

Lab sheet 07**Submission Due: 03.07.2024****Title:** Getting and Passing Data from HTML to PHP Using Ajax.**Aims:**

- To understand the concept of Ajax and its benefits.
- To learn how to make Ajax request with JQuery.
- To learn how to process Ajax request in PHP

Tasks:

- Import JQuery
- Implement AJAX request with JQuery
- Perform text field validation
- Process data in PHP

Activities :

1. Import JQuery

```
<script  
src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js  
></script>
```

2. Implement AJAX request with JQuery
 - I. POST request

```
$.ajax({  
    url: "add_student.php",  
    type: "POST",  
    data: {  
        name: name,  
        gpa: gpa  
    },  
    success: function(response) {  
        try {  
            var data = JSON.parse(response);  
  
            if (data.success) {  
                $("#studentForm").trigger("reset");  
                console.log("Student added  
successfully!");  
  
                if (data.students) {  
                    $("#studentTable  
tbody").empty();
```

```

                                for (var i = 0; i <
data.students.length; i++) {
                                var student =
                                data.students[i];
                                var tableRow = "<tr>";
                                tableRow += "<td>" +
                                student.id + "</td>";
                                tableRow += "<td>" +
                                student.name + "</td>";
                                tableRow += "<td>" +
                                student.gpa + "</td>";
                                tableRow += "</tr>";
                                $("#studentTable
tbody").append(tableRow);
                                }
                                } else {
                                alert("Error adding student: " +
data.message);
                                }
                                } catch (error) {
                                console.error("Error parsing
response:", error);
                                alert("An error occurred while adding
student. Please check your server logs.");
                                }
                                },
                                error: function (jqXHR, textStatus,
errorThrown) {
                                console.error("AJAX Error:", textStatus,
errorThrown);
                                alert("An error occurred while adding
student.");
                                }
                                });

```

II. GET request

```

$.ajax({
    url: "get_students.php",
    type: "GET",
    success: function(response) {
        try {
            var data = JSON.parse(response);

            if (data.success && data.students &&
Array.isArray(data.students)) {
                $("#studentTable tbody").empty();

```

```

        for (var i = 0; i < data.students.length;
i++) {
            var student = data.students[i];
            var tableRow = "<tr>";
            tableRow += "<td>" + student.id +
"</td>";
            tableRow += "<td>" + student.name +
"</td>";
            tableRow += "<td>" + student.gpa +
"</td>";
            tableRow += "</tr>";
            $("#studentTable
tbody").append(tableRow);
        }
    } else {
        console.warn("Unexpected response format:
students array not found.");
    }
} catch (error) {
    console.error("Error parsing response:",
error);
    alert("An error occurred while retrieving
student data.");
}
},
error: function(jqXHR, textStatus, errorThrown) {
    console.error("AJAX Error:", textStatus,
errorThrown);
    alert("An error occurred while retrieving student
data.");
}
});

```

3. Perform text field validation

```

var name = $("#name").val();
var gpa = parseFloat($("#gpa").val());

if (name.trim() === "") {
    alert("Please enter a name.");
    return;
}
if (isNaN(gpa) || gpa < 0 || gpa > 4) {
    alert("Please enter a valid GPA between 0 and
4.");
    return;
}

```

4. Process data in PHP

I. get_stidents.php

```
if ($_SERVER['REQUEST_METHOD'] === 'GET') {  
  
    include ('DatabaseConnection.php');  
    include ('Student.php');  
    $db = new DatabaseConnection("localhost", "root", "",  
"faculty");  
    $connection = $db->getConnection();  
  
    $student = new Student($connection);  
  
    $studentData = $student->getAllStudent();  
  
    $response = array();  
    if ($studentData) {  
        $response['success'] = true;  
        $response['students'] = $studentData;  
    } else {  
        $response['success'] = false;  
        $response['message'] = "Error creating student";  
    }  
  
    echo json_encode($response);  
} else {  
    echo "Invalid request."  
}
```

II. add_students.php

```
if ($_SERVER['REQUEST_METHOD'] === 'POST') {  
    $name = $_POST['name'];  
    $gpa = floatval($_POST['gpa']);  
  
    include ('DatabaseConnection.php');  
    include ('Student.php');  
    $db = new DatabaseConnection("localhost", "root", "",  
"faculty");  
    $connection = $db->getConnection();  
  
    $student = new Student($connection);  
    $student->setName($name);  
    $student->setGpa($gpa);  
  
    $result = $student->create();  
}
```

```
$response = array();  
if ($result) {  
    $response['success'] = true;  
    $studentData = $student->getAllStudent();  
    $response['students'] = $studentData;  
} else {  
    $response['success'] = false;  
    $response['message'] = "Error creating student";  
}  
  
echo json_encode($response);  
} else {  
    echo "Invalid request."  
}
```

Discussion:

- Advantages of using AJAX with jQuery and PHP